EXPERIMENT 13: Fibonacci sequence using module

A]ALGORITHM:

Step1: **Start**

Step 2: **Create Module (fibonacci_module.py):**
a. Define a function generate_fibonacci_sequence(max_value)
b. If max_value < 0, return empty list
c. Initialize:

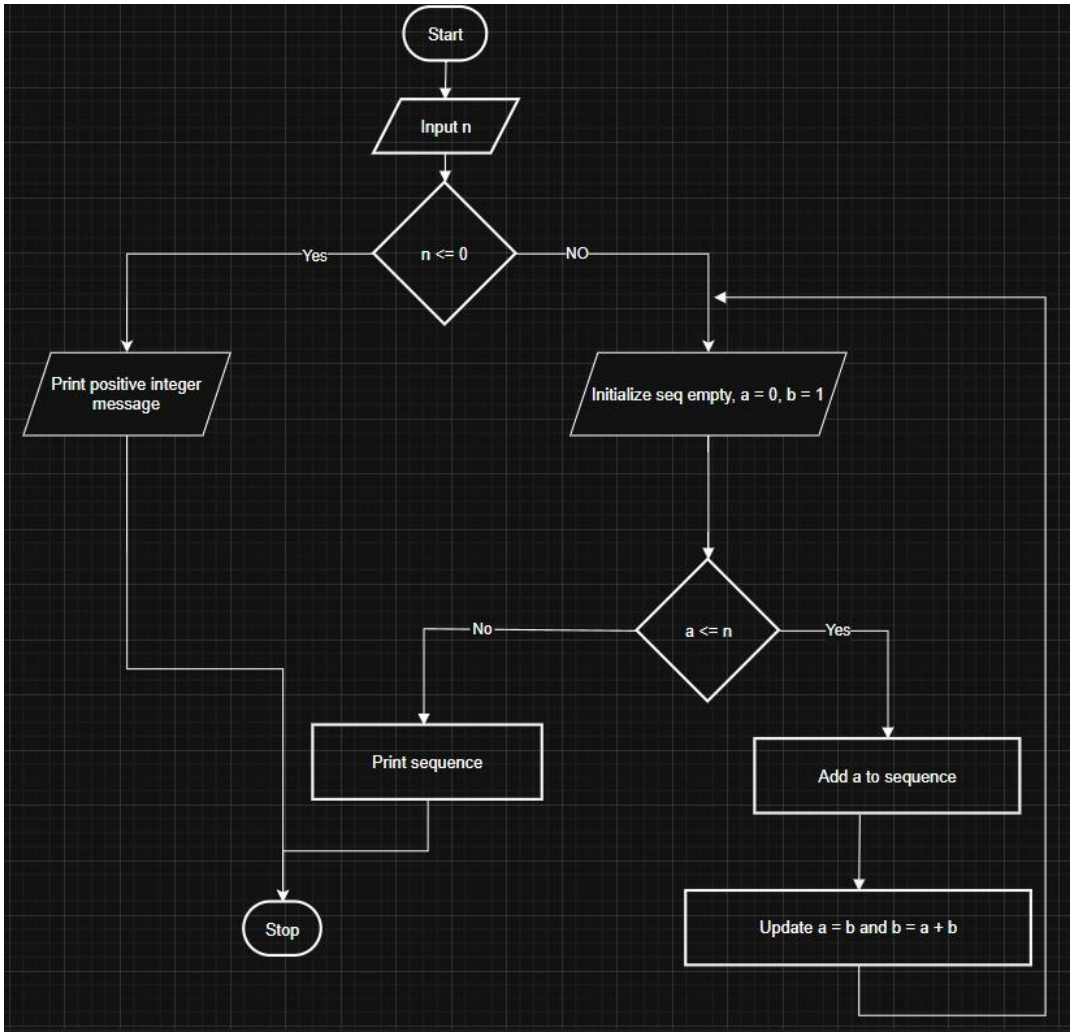
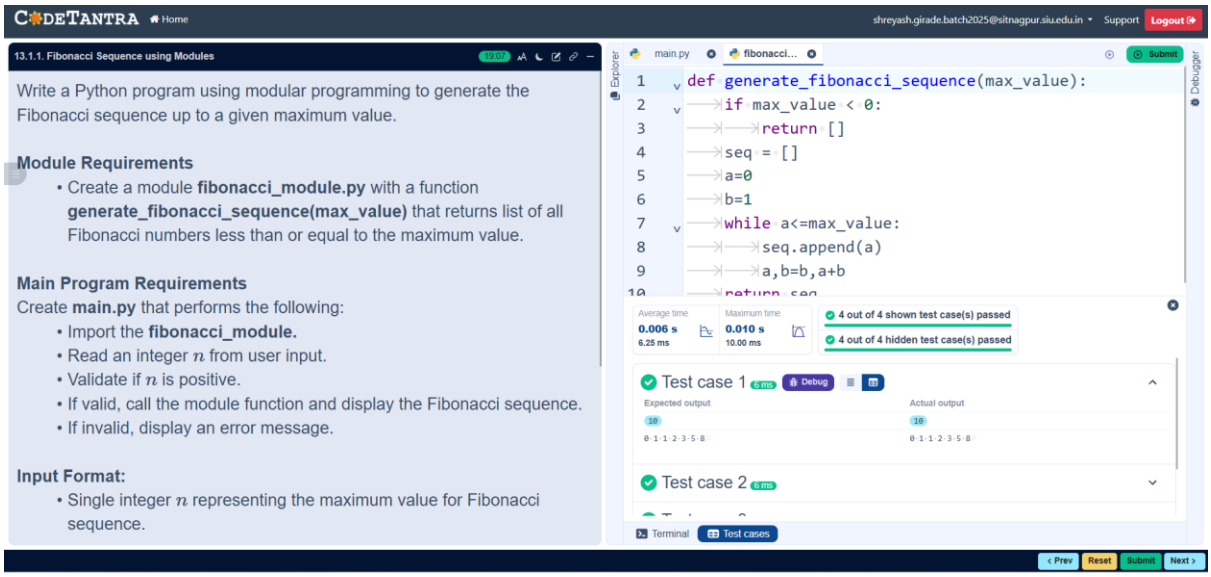
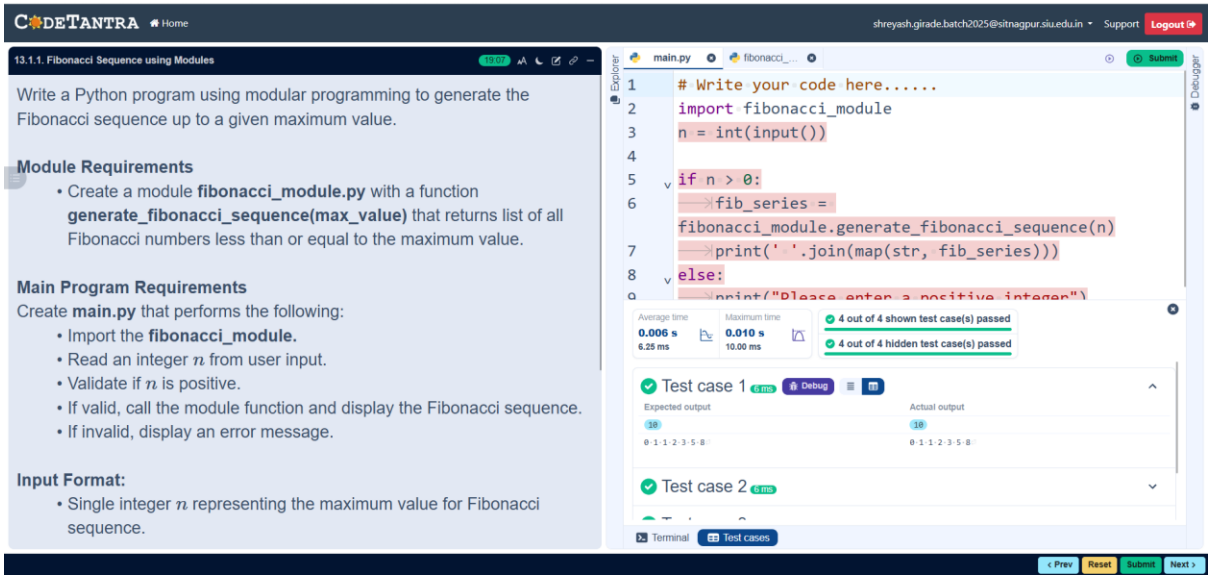
- a = 0
- b = 1
- seq = []
  d. Repeat while a ≤ max_value:
- Add a to seq
- Update values:
  a = b
  b = a + b (use proper swapping: a, b = b, a + b)
  e. Return seq

Step 3: **Main Program (main.py):**
a. Import the module fibonacci_module
b. Read integer input n
c. Check if n > 0:

- If **true**:
  - Call function generate_fibonacci_sequence(n)
  - Store result in fib_series
  - Print all elements of fib_series separated by space
- If **false**:
  - Print "Please enter a positive integer"

Step 4: **End**

B]FLOWCHART:


Start
Input n
n <= 0
Yes
NO
Print positive integer message
Initialize seq empty, a = 0, b = 1
a <= n
No
Yes
Print sequence
Add a to sequence
Stop
Update a = b and b = a + b

C] CODETANTRA: